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Python Modules

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Python modules:

- A module is a piece of software that has a specific functionality.
- A Python module is a file that contains Python code.
- Modules in Python provides us the flexibility to organize the code in a logical way.
- To use the functionality of one module into another, we must have to import the specific module.
- python code file saved with the extension (.py) is treated as the module.

We need to load the module in our python code to use its functionality.

Python provides two types of statements as defined below.

1. The import statement
2. The from-import statement

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```python
#place the code in the calculation.py
def summation(a,b):
    return a+b
def multiplication(a,b):
    return a*b;
def divide(a,b):
    return a/b;
```

```python
#main.py
from calculation import summation
#it will import only the summation() from calculation.py
a = int(input("Enter the first number"))
b = int(input("Enter the second number"))
 print("Sum = ",summation(a,b))
#we do not need to specify the module name while accessing summation()
```

We can also import all the attributes from a module by using *.

Consider the following syntax.

```
from &lt;module&gt; import *
```

Renaming a module:

```
#the module calculation of previous example is imported in this example as cal.
import calculation as cal;
a = int(input("Enter a?"));
b = int(input("Enter b?"));
print("Sum = ",cal.summation(a,b))
```

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Python packages

The packages in python facilitate the developer with the application development environment by providing a hierarchical directory structure where a package contains sub-packages, modules, and sub-modules.

The packages are used to categorize the application level code efficiently.

Let's create a package named Employees in your home directory.
Consider the following steps.

1. Create a directory with name Employees on path /home.

2. Create a python source file with name ITEmployees.py on the path /home/Employees.

ITEmployees.py

```
def getITNames():
    List = ["John", "David", "Nick",    "Martin"]
    return List;
```

Similarly, create one more python file with name BPOEmployees.py and create a function getBPONames().

To make this directory a package, we need to include one more file here, that is __init__.py which contains the import statements of the modules defined in this directory.

__init__.py

```
from ITEmployees import getITNames
from BPOEmployees import getBPONames
```

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Let's create a simple python source file at our home directory (/home) which uses the modules defined in this package.

Test.py

```
import Employees
print(Employees.getNames())
```

Output:

```
['John', 'David', 'Nick', 'Martin']
```